

## **Lab 3 Report**

**Jose Montalvo**

## Part 1



Figure 1:

Blue is Roll from Acceleration.

Orange is Pitch from Acceleration.

Pink is Roll from Complimentary.

Light Blue is pitch complimentary.

The Comp filters take the gross accelerometer data and the drifting gyro data and blends them together using a weighted average. my  $P = 0.75$

## Part 2



Blue is the Unfiltered data.

Orange is the Butters filter on the data. Green is the median filter on the data.

Yellow is the moving average filter on the data.

Pink is the Peak Detection on the Butters Filtered data.

The three filters I am using are Buttersworth, Median and the moving average. I seem to get more consistent values with the Buttersworth filtered data than with the other filters. I was running a  $f_{\text{sampling}} = 10\text{Hz}$  and a  $f_{\text{cutoff}} = 2\text{Hz}$

## Part 3

List online sources and collaborators.

No additional online sources outside of the lab links. I did not work with another student.

## Part 4

Self-evaluation:

- (a) What did you expect to learn? I hoped to learn about accelerometer data collection
- (b) Rate class participation and needed adjustments. I feel like I did a great job staying on top of the assignments.
- (c) Grade yourself (0–4) and explain. I would give myself a 4. I had the majority of the lab done by monday of this week.
- (d) What did you actually learn? I learned that there is a lot of noise from sensors and that being able to use peak detection is more useful when you feed already filtered data. You can use unfiltered data but then you would also be picking up the noise in the system.

## Code

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%CODE HERE

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